P510/2

PHYSICS

Paper 2

(Practical)

$3\frac{1}{4}$ hours

TESO SECONDARY SCHOOLS MOCK EXAMINATIONS ASSOCIATION (TESSMEA)

Uganda Advanced Certificate of Education

PHYSICS PRACTICAL

Paper 2

(PRINCIPAL SUBJECT)

3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of **two** *compulsory items.*

Use blue or black ink.

Any work done in pencil except graphs and diagrams will not be scored.

Scores are awarded mainly for a clear record of the observations actually made, for their suitability, accuracy and for the use made of them.

Details on the question paper should not be repeated in the responses.

Graph paper is provided.

Silent non-programmable scientific calculators may be used.

Turn Over

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Item 1

A local technician was contracted for routine maintenance and repairs on lighting devices and bulbs, which operate automatically by use of the temperature sensors. The temperature sensors rely more on the resistivity of the wires used in the connection and potential dividers.

The technician realized that some spoilt devices required a resistance wire whose resistivity is about $5x10^{-7}$ Ωm to $2.0x10^{-6}$ Ωm for correct repairs, according to the specification he saw from the devices. He therefore, ordered for that specific brand of wires to use. The wires were supplied without any specifications and he wants to first know its resistivity before any use, since it will affect the brightness and the work of the devices.

However, the technician could not test the resistivity of the supplied wire, since the machine he uses for testing the resistivity of wires (resistivity meter) was spoilt. He therefore consults your teacher for help with a sample of the supplied wire, labelled X. The teacher plans to carry out the investigation using two different methods and get the average value so that the results can be trusted.

In method I, the teacher intended to connect, once, the test wire in question labelled X, a voltmeter and an ammeter. In the second method, the teacher intended to use a meter bridge connection, balancing the test wire X of varying lengths against a 5Ω standard resistance and analyze the results graphically. The teacher has requested you as one of the students to perform the investigation since he was engaged.

Task:

Design and carry out a scientific investigation, by using the two methods, to determine the suitability of the supplied wire.

Item 2

A Science club was designing highly sensitive beam balances for experiments to be carried out under Micro-gravity conditions in the International Space Station. The spring needed to achieve these conditions must strictly have a force constant in the range of $20Nm^{-1}$ to $50Nm^{-1}$ and an effective mass that lies between 5g to 20g inclusive for sensitivity and accuracy of the experiments carried out using such balances.

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The Science club ordered and received springs which they expect to use in the assembling of the beam balances. However, they need to first test for their suitability since this would affect the reliability of the results from the experiments. But, since the springs were quite many, the club chairperson approached your teacher requesting for help. The teacher advised that they could go and use a group of learners of S.6 class, give them instructions and they carry out the investigation under supervision.

You are provided with the spring and the other apparatus required for the investigation.

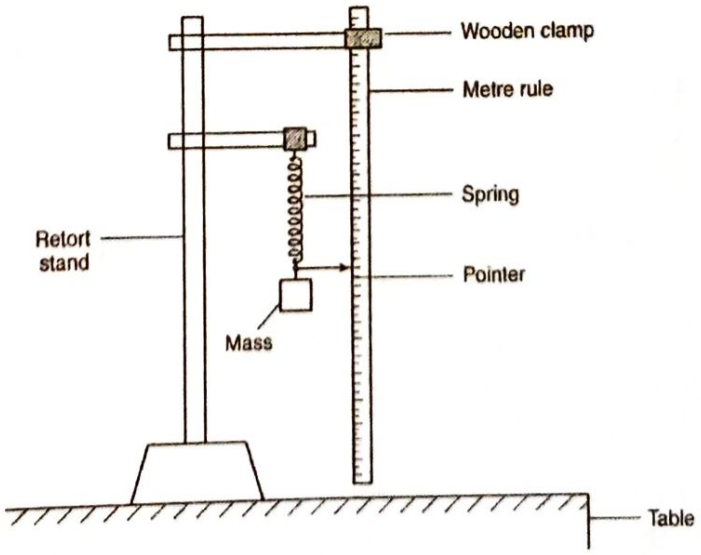
(**HINT**; the period of oscillation of a loaded spring is related to the total mass as $T = 2\pi\sqrt{\frac{M+m}{k}}$, where M is the mass loaded, k is force constant of the spring and m is effective mass of the spring.

Task:

Follow the instructions provided and carry out the investigation and give feedback to the club.

Method 1

(a) Suspend the spring in the retort stand as shown in the figure below.



(b) Attach the pointer at the lower end of the spring.

(c) Read and record the initial pointer position P_o from the meter rule in meters.

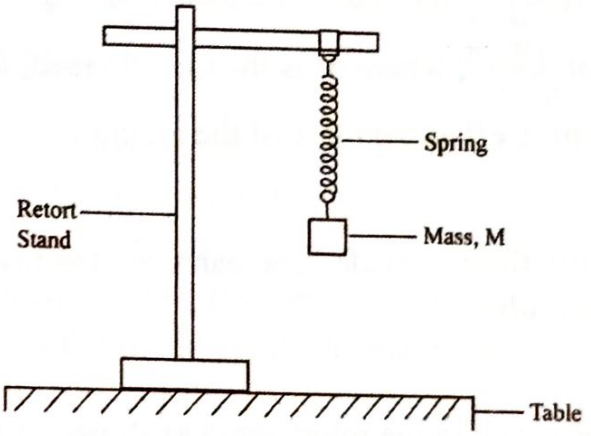
(d) Suspend the mass M_1 = 0.200Kg at the free end of the spring.

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(e)Read and record the final pointer position P_1 from the meter rule in meters
(f) Determine the extension $e_1$of the spring.
(g)Obtain the value of K' from appropriate expression
(h)Repeat the steps (d) to (f) for the value of $M_2 = 0.400$Kg each time noting P_2 and value of e_2 respectively, then obtain the value of K''
(i) Determine average value of force constant K_1 from $K_1 = \frac{1}{2}(K' + K'')$.

Method 2

(a)Suspend the spring on the retort stand as shown in the diagram below.



(b)Attach the mass $M = 0.100kg$ on the free end of the spring.
(c)Give a spring a small vertical downward displacement and release it to oscillate.
(d)Read and record the time t for 20 oscillations.
(e)Determine the period T for one complete oscillation.
(f) Repeat the steps (b) to (e) for the values of
$M = 0.200, 0.300, 0.400, 0.500$ *and* $0.600kg$
(g)Tabulate your results including values of T^2
(h)Plot a graph of T^2 against **M.**
(i) Determine the slope, S of the graph and the intercept C, on the $T^2 - axis$
(j) Obtain the value of K_2 and m from the appropriate expression.
(k)Find the average value of the force constant **K** from $K = \frac{1}{2}(K_1 + K_2)$.
(l) Give a feed-back to the club members based of your findings.

END

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